TDRA HACKATHON 2026 · CHALLENGE #2 · GCAA

DABAB

An explainable AI copilot that helps air traffic control recover from fog and severe weather. The AI recommends. The human decides.

[Team name] · [Member 1] · [Member 2]

01 · THE HOOK

Saturday, 3 January 2026. 07:04. Fog closes in over Dubai International.

Visibility drops to 193 metres. In one hour, landings fall from 28 to 15. Dozens of aircraft are in the air, and controllers must decide, flight by flight, who holds, who diverts, and where.

Sources: OMDB METAR (Iowa Environmental Mesonet); Dubai Airports flight information, Data.Dubai

02 · THE CHALLENGE

Weather decisions are made fast, by hand, with three systems that do not talk

Minutes, not hours

Holding, diversion and stand choices for dozens of flights at once, under stress.

Disconnected views

Air navigation, the airport and airlines each see only their part of the picture.

Hidden trade-offs

Less delay can mean more fuel burned in holding. The cost of each choice is not visible.

03 · THE EVIDENCE

Bad weather at DXB multiplies arrival delays

Average arrival delay in fog

86 min

vs 11 min on normal days

Weather at DXB	Events	Delay (min)	>15 min late
Fog	5	86 vs 11	83% vs 28%
Thunderstorm	2	102 vs 15	81% vs 33%
Dust / sand	42	10 vs 8	27% vs 23%

Event vs same hours on normal days within ±14 days. Small samples: observed averages, not statistical proof.

Data: Dubai Airports arrivals (282,705 flights, May 2025 – Nov 2026), Data.Dubai; OMDB METAR

04 · THE POSSIBILITY

Imagine the same morning, with three ready plans on screen before the fog peaks

Seconds

Ranked recovery plans the moment visibility drops

One picture

Air navigation, airport and airlines see the same plan

Every weather

Fog first; thunderstorms and dust with the same engine

05 · THE SOLUTION

The controller’s journey with DABAB

1

Alert

Fog detected from METAR. Landing rate cut.

2

Priority

Controller picks: less delay, less fuel, fewer diversions or connections.

3

Plans

Three plans with delay, fuel, CO_2, diversions and risk.

4

Explain

Why this plan, what it costs, why not the others. Arabic and English.

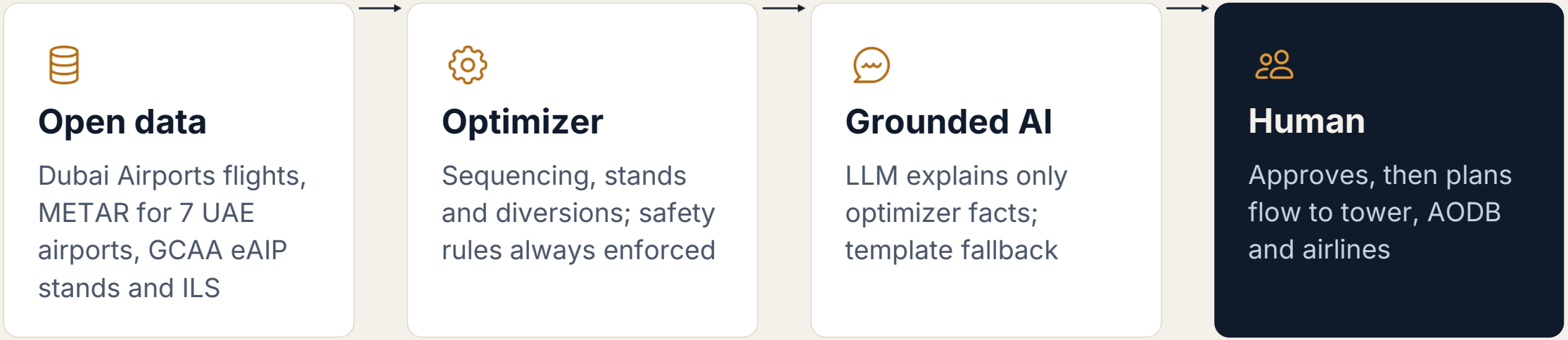
5

Human decides

Approve, modify or reject. Sent to tower, airport and airlines. Logged.

06 · THE TECHNOLOGY

Four layers: the AI recommends and explains, the human decides



Built as a layer on A-CDM, using the SWIM/FIXM standard interfaces between air navigation and airport systems.

07 · THE BUSINESS MODEL

Create, deliver and sustain value

Create

Customers: GCAA and the air navigation provider (dans), then airports and airlines.

Users: controllers, airport and airline operations centres.

Deliver

GCAA-led pilot at DXB, running on top of A-CDM.

Partners: GCAA, dans, Dubai Airports, NCM, Digital Dubai, airlines.

Sustain

Revenue: annual licence per airport (SaaS) plus integration and training.

Savings: less holding fuel, fewer diversions, fewer missed connections, less CO_2.

08 · THE TEAM

Two builders who already have it running on real data

Ali Eissa

Engine and evidence

Data pipeline, optimizer, fog model, grounded explanations.

Mohammed Hishaam

Product and pitch

ATC decision panel, cockpit view, business model, pitch.

Built so far: 282,705 real arrivals analysed, 220 DXB stands mapped from GCAA charts, a working engine and ATC panel.

09 · THE PATHWAY

90 days: from prototype to a tested system

Days 1–30

Rebuild to industry standard

- Rewrite the prototype with code review, automated tests and version control
- Security: access roles, encryption, signed audit log, UAE data residency
- Follow aviation software assurance practice (DO-278A / ED-109A)

Milestone: security and code review passed

Days 31–60

Train the AI, build the endpoint

- Train the fog model on more years of METAR and flight data
- Tune plan ranking from controllers' approve, modify and reject choices
- Secure API that sends only approved plans to other systems (SWIM/FIXM)

Milestone: API works end to end in a test sandbox

Days 61–90

Test it, simulated then real

- Replay every past fog and storm day; measure delay and fuel saved
- Mock fog drills with controllers in a simulator
- If GCAA approves: live trial at one airport, suggestions only

Milestone: go / no-go with GCAA

10 · THE CLOSING STATEMENT

When the weather changes, nobody should have to guess.

DABAB: the AI recommends, explains and is accountable. The human decides.